

Practice Quiz: Agents

Part A: Multiple Choice

1. In Active Inference, a crocheter is best described as: A) A passive receiver of pattern instructions B) A free energy minimizing agent who acts to reduce the gap between expectations and observations C) A random stitch generator D) An entity that only responds to external commands
2. The crochet pattern you hold in your mind while working functions as: A) A decorative accessory B) A generative model that predicts what the fabric should look like at each stage C) An external state that has no influence on your actions D) A fixed rule that cannot be updated
3. When a crocheter counts stitches at the end of a row and finds the correct number, this is an example of: A) Prediction error increasing B) The Markov blanket dissolving C) Belief updating — confirming the generative model's prediction with sensory evidence D) Free energy maximization
4. Frogging (ripping back stitches to correct an error) is best understood in Active Inference as: A) A random action with no purpose B) Active inference — changing the world to match the generative model C) Giving up on the project entirely D) Increasing prediction error intentionally
5. An experienced crocheter can "feel" a mistake before seeing it. This suggests their generative model has: A) No predictive power B) Lower precision than a beginner's C) High precision — it generates detailed expectations that detect small deviations D) Been completely replaced by muscle memory
6. When a crocheter decides a small irregularity is "a design feature, not a mistake," they are: A) Failing as an agent B) Updating their generative model to accommodate the observation, thereby reducing free energy C) Increasing their prediction error D) Ignoring Active Inference principles entirely

7. A crocheter tries a completely new stitch for the first time. Compared to working a familiar stitch, free energy is likely: A) Lower, because novelty reduces surprise B) Higher, because the generative model is sparse and predictions are less accurate C) Exactly the same regardless of experience D) Zero, because the pattern provides all necessary information

Part B: Short Answer

1. Describe the two fundamental strategies a crocheter-as-agent has for reducing free energy when they discover an error, and give a concrete crochet example of each.
2. Explain how a beginner crocheter's generative model differs from an expert's. What specific aspects of crochet knowledge become encoded in the model over time?
3. You are making a blanket and realize after 30 rows that your gauge has been slightly off, making the blanket narrower than intended. Describe your options as an Active Inference agent and the trade-offs involved in each.